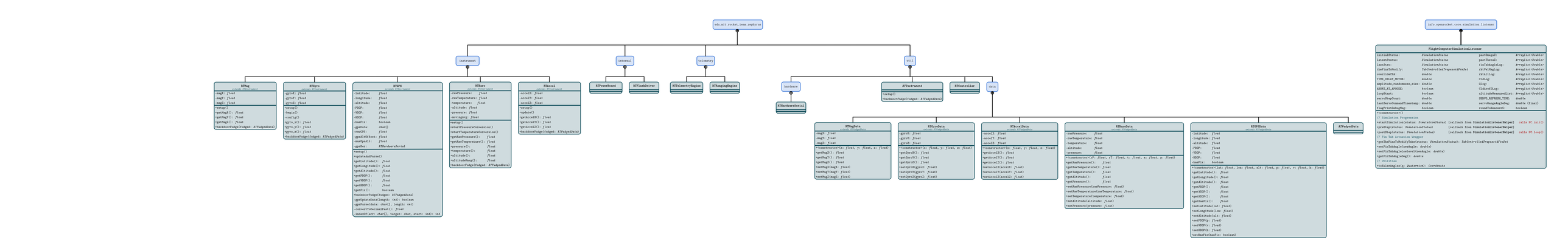

edu.mit.rocket_team.zephyrus
instrument
RTMag
extends RTInstrument
-magX: float
-magY: float
-magZ: float
+setup()
+getMagX(): float
+getMagY(): float
+getMagZ(): float
+backdoorFudge(fudged: RTFudgedData)
RTGyro
extends RTInstrument
-gyroX: float
-gyroY: float
-gyroZ: float
+setup()
-begin()
-config()
+gyro_x(): float
+gyro_y(): float
+gyro_z(): float
+backdoorFudge(fudged: RTFudgedData)
RTGPS
extends RTInstrument
-latitude: float
-longitude: float
-altitude: float
-PDOP: float
-VDOP: float
-HDOP: float
-hasFix: boolean
-gpsData: char[]
-rawGPS: float
-gpsAltOffset: float
-maxGpsAlt: float
-gpsSer: RTHardwareSerial
+setup()
+updateAndParse()
+getLatitude(): float
+getLongitude(): float
+getAltitude(): float
+getPDOP(): float
+getVDOP(): float
+getHDOP(): float
+getFix(): boolean
+backdoorFudge(fudged: RTFudgedData)
-gpsUpdateData(length: int): boolean
-gpsParse(data: char[], length: int)
-convertToDecimalFast(): float
-indexOf(arr: char[], target: char, start: int): int
RTBaro
extends RTInstrument
-rawPressure: float
-rawTemperature: float
-temperature: float
-altitude: float
-pressure: float
-movingAvg: float
+setup()
+startPressureConversion()
+startTemperatureConversion()
+getRawPressure(): float
+getRawTemperature(): float
+pressure(): float
+temperature(): float
+altitude(): float
+altitudeMavg(): float
+backdoorFudge(fudged: RTFudgedData)
RTAccel
extends RTInstrument
-accelX: float
-accelY: float
-accelZ: float
+setup()
+update()
+getAccelX(): float
+getAccelY(): float
+getAccelZ(): float
+backdoorFudge(fudged: RTFudgedData)
internal
RTPowerBoard
RTFlashDriver
telemetry
RTTelemetryEngine
RTRangingEngine
util
hardware
RTHardwareSerial
RTInstrument
+setup()
+backdoorFudge(fudged: RTFudgedData)
RTController
data
RTMagData
extends RTFudgedData
-magX: float
-magY: float
-magZ: float
+<constructor>(x: float, y: float, z: float)
+getMagX(): float
+getMagY(): float
+getMagZ(): float
+setMagX(magX: float)
+setMagY(magY: float)
+setMagZ(magZ: float)
RTGyroData
extends RTFudgedData
-gyroX: float
-gyroY: float
-gyroZ: float
+<constructor>(x: float, y: float, z: float)
+getGyroX(): float
+getGyroY(): float
+getGyroZ(): float
+setGyroX(gyroX: float)
+setGyroY(gyroY: float)
+setGyroZ(gyroZ: float)
RTAccelData
extends RTFudgedData
-accelX: float
-accelY: float
-accelZ: float
+<constructor>(x: float, y: float, z: float)
+getAccelX(): float
+getAccelY(): float
+getAccelZ(): float
+setAccelX(accelX: float)
+setAccelY(accelY: float)
+setAccelZ(accelZ: float)
RTBaroData
extends RTFudgedData
-rawPressure: float
-rawTemperature: float
-temperature: float
-altitude: float
-pressure: float
+<constructor>(rP: float, rT: float, t: float, a: float, p: float)
+getRawPressure(): float
+getRawTemperature(): float
+getTemperature(): float
+getAltitude(): float
+getPressure(): float
+setRawPressure(rawPressure: float)
+setRawTemperature(rawTemperature: float)
+setTemperature(temperature: float)
+setAltitude(altitude: float)
+setPressure(pressure: float)
RTGPSData
extends RTFudgedData
-latitude: float
-longitude: float
-altitude: float
-PDOP: float
-VDOP: float
-HDOP: float
-hasFix: boolean
+<constructor>(lat: float, lon: float, alt: float, p: float, v: float, h: float)
+getLatitude(): float
+getLongitude(): float
+getAltitude(): float
+getPDOP(): float
+getVDOP(): float
+getHDOP(): float
+getHasFix(): float
+setLatitude(lat: float)
+setLongitude(lon: float)
+setAltitude(alt: float)
+setPDOP(p: float)
+setVDOP(v: float)
+setHDOP(h: float)
+setHasFix(hasFix: boolean)
RTFudgedData
info.openrocket.core.simulation.listener
FlightComputerSimulationListener
initialStatus: SimulationStatus
latestStatus: SimulationStatus
lastStat: SimulationStatus
theFinsToModify: TabControlledTrapezoidFinSet
overrideCNA: double
TIME_DELAY_MOTOR: double
amplitude_randomness_size: double
ABORT_AT_APOGEE: boolean
loopStart: boolean
servoStepCount: double
lastServoCommandTimestamp: double
flagPrintDebugMsg: boolean
pastOmegaZ: ArrayList<Double>
pastThetaZ: ArrayList<Double>
finTabAngleLog: ArrayList<Double>
rktVelMagLog: ArrayList<Double>
rktAltLog: ArrayList<Double>
CldLog: ArrayList<Double>
Qlog: ArrayList<Double>
CldArefDLog: ArrayList<Double>
altitudeMeasuredList: ArrayList<Double>
SERVO_REFRESH_TIME: double
servoRangeAngleDeg: double {final}
roundToNearest5: boolean
+<constructor>()
// Simulation Progression
+startSimulation(status: SimulationStatus) [callback from SimulationListenerHelper] calls FC.init()
+preStep(status: SimulationStatus) [callback from SimulationListenerHelper]
+postStep(status: SimulationStatus) [callback from SimulationListenerHelper] calls FC.loop()
// Fin Tab Actuation Wrapper
+getTheFinsToModifyTabs(status: SimulationStatus): TabControlledTrapezoidFinSet
+setFinTabAngle(newAngle: double)
+setFinTabAngleLowlevel(newAngle: double)
+getFinTabAngleDeg(): double
// Utilities
+toEulerAngles(q: Quaternion): Coordinate